

Secure Exit

CO3-AT3-Assignment

← Back



SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1Banking Transaction System –
Exception Handling

10 marks

**Q2**Hospital Patient Registration –
User-Defined Exception

10 marks

**Q3**E-Commerce Order Processing
– Built-in Exceptions

10 marks

**Q4**Hospital Emergency System –
Thread Priority

10 marks

**Q5**Online Food Delivery – Multiple
Threads and Execution Order

10 marks



5/5 answered

Q4 Hospital Emergency System – Thread Priority

10 marks

A hospital application creates threads for Emergency Patient Processing, General Patient Processing, and Billing Processing.

Develop a Java program that assigns different priorities to these threads. Run the program several times and analyze whether the emergency thread is guaranteed to execute first.

Then modify the design if the hospital requires a strict execution sequence rather than merely expressing scheduling preference.

Your answer must contain Java code and an execution analysis based on the actual thread behavior.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus

Emergency, General, Billing All threads execute; priority does not guarantee a fixed output order Normal execution

Emergency workload > General workload Emergency thread receives a higher priority but exact ordering is not guaranteed Concurrency / design

Strict emergency-before-billing requirement Use explicit coordination rather than relying only on priority Exception / boundary

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         EmThread em=new EmThread();
5         GeneralThread g= new GeneralThread();
6         BThread b=new BThread();
```

```
7         em.start();
8         g.start();
9         b.start();
10    }
11 }class EmThread extends Thread{
12     public void run(){
13         System.out.println("Emergency Patient processing");
14     }
15 }
16 class GeneralThread extends Thread{
17     public void run(){
18         System.out.println("General patient processing");
19     }
20 }
21 class BThread extends Thread{
22     public void run(){
23         System.out.println("Billing processing");
24     }
25 }
```

Input

stdin input

Output

```
Emergency Patient processing  
General patient processing  
Billing processing
```

✓ Submitted

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